Back

Module 3 Quiz: Mining Vector and Matrix Data
Graded Assignment • 30 min

Due Nov 2, 11:59 PM IST

## Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

**1.** What is the advantage of using vectors and matrices for data representation? 1 / 1 point

- (•) They allow for efficient computation and analysis of multi-dimensional data.
- ( ) They can only handle categorical data.
- ( ) They require a fixed data structure like dictionaries.
- ( ) They ignore the quantity of items in the dataset.

**Nice work**

Correct! Vectors and matrices facilitate efficient computation and analysis of data with multiple dimensions.

**2.** What does the direction of a vector in a coordinate system represent? 1 / 1 point

- ( ) The length of the vector.
- ( ) The number of dimensions in the system.
- (•) The identity of the vector in that system.
- ( ) The type of data the vector represents.

**Nice work**

Correct! The direction of a vector in a coordinate system represents its identity based on the values in each dimension.

**3.** What is the result of adding two vectors in a vector space? 1 / 1 point

- ( ) A vector with the same direction as the original vectors.
- ( ) A matrix with increased dimensions.
- ( ) A scalar value representing the sum of all elements.
- (•) A new vector with a new direction and magnitude.

**Nice work**

Correct! Adding two vectors results in a new vector with its own direction and magnitude.

**4.** In a matrix, suppose each column corresponds to an attribute, what does each row vector represent? 1 / 1 point

- (•) A data object with values corresponding to specific dimensions.
- ( ) A binary vector with only 0s and 1s.
- ( ) A column vector transpose.
- ( ) An attribute across all data objects.

**Nice work**

Correct! Each row vector represents a data object with p values, each corresponding to a specific dimension.

**5.** What does each column vector in a matrix represent? 1 / 1 point

- ( ) A transposed row vector.
- (•) An attribute across all data objects.
- ( ) A single data object.
- ( ) A binary vector with only 0s and 1s.

**Nice work**

Correct! Each column vector represents an attribute across all data objects.

**6.** What is a potential problem with using the dot product as a similarity measure? 1 / 1 point

- ( ) It cannot handle real numbers.
- ( ) It requires vectors to be of different lengths.
- (•) It can indicate a vector is more similar to another vector than to itself.
- ( ) It may result in a zero value for similarity.

**Nice work**

Correct! The dot product can sometimes indicate that a vector is more similar to another vector than to itself, which is counterintuitive.

**7.** What does a positive product of corresponding dimensions in two vectors indicate? 1 / 1 point

- ( ) One value is zero.
- ( ) The vectors are identical.
- ( ) The vectors are of different lengths.
- (•) Both values have the same sign.

**Nice work**

Correct! A positive product indicates that both values in the corresponding dimensions have the same sign, contributing positively to similarity.

**8.** What is a key property of both Manhattan and Euclidean distances? 1 / 1 point

- (•) They both satisfy the triangle inequality.
- ( ) They are both affected by vector direction, but not by the magnitude.
- ( ) They are both used to calculate angles between vectors.
- ( ) They are both non-symmetric.

**Nice work**

Correct! Both Manhattan and Euclidean distances satisfy the triangle inequality, ensuring the direct path is the shortest.

**9.** What is the Euclidean distance primarily used to measure? 1 / 1 point

- ( ) The similarity between two vectors.
- (•) The straight-line distance between two points.
- ( ) The sum of absolute differences between vector components.
- ( ) The angle between two vectors.

**Nice work**

Correct! Euclidean distance measures the straight-line distance between two points in a vector space.

**10.** What is a limitation of using the dot product for measuring similarity? 1 / 1 point

- ( ) It calculates the angle between vectors.
- ( ) It is not affected by vector magnitude.
- ( ) It measures the distance between vectors.
- (•) It can exaggerate similarity for vectors with larger values.

**Nice work**

Correct! The dot product can exaggerate similarity for vectors with larger values due to multiplication of values across dimensions.

**11.** What does a cosine similarity of 0 between two vectors imply? 1 / 1 point

- ( ) The vectors are highly correlated.
- (•) The vectors are orthogonal and unrelated.
- ( ) The vectors are identical.
- ( ) The vectors are opposite in direction.

**Nice work**

Correct! A cosine similarity of 0 indicates that the vectors are orthogonal and unrelated.

**12.** What does a negative cosine similarity indicate about two vectors? 1 / 1 point

- ( ) The vectors are orthogonal.
- ( ) The vectors are unrelated.
- (•) The vectors are opposite in direction.
- ( ) The vectors are identical.

**Nice work**

Correct! A negative cosine similarity indicates that the vectors are opposite in direction.

**13.** How is the Pearson coefficient related to cosine similarity? 1 / 1 point

- (•) It is a cosine similarity with the mean removed from each vector.
- ( ) It is unrelated to cosine similarity.
- ( ) It is the square of the cosine similarity.
- ( ) It is a cosine similarity with added dimensions.

**Nice work**

Correct! The Pearson coefficient is essentially cosine similarity after subtracting the mean of the dimensions from each vector.

**14.** What does the Manhattan distance measure between two vectors? 1 / 1 point

- ( ) The angle between two vectors.
- ( ) The square root of the sum of squared differences.
- ( ) The product of corresponding values in corresponding dimensions.
- (•) The sum of the absolute differences across dimensions.

**Nice work**

Correct! Manhattan distance is the sum of the absolute differences across dimensions.

**15.** Which similarity metric is recommended for homogeneous vector dimensions, such as user ratings? 1 / 1 point

- ( ) Cosine similarity
- ( ) Euclidean distance
- (•) Pearson's coefficient
- ( ) Manhattan distance

**Nice work**

Correct! Pearson's coefficient is suitable for homogeneous data like user ratings.

**16.** In the context of eigendecomposition, what does the diagonal matrix Λ represent? 1 / 1 point

- ( ) The eigenvectors of the matrix.
- ( ) The inverse of the eigenvectors matrix.
- ( ) The determinant of the matrix.
- (•) The eigenvalues of the matrix.

**Nice work**

Correct! The diagonal matrix Λ contains the eigenvalues of the matrix.

**17.** Which of the following statements is true about eigendecomposition? 1 / 1 point

- (•) It is a method to express a matrix in terms of its eigenvalues and eigenvectors.
- ( ) It is used to find the inverse of a matrix.
- ( ) It can be applied to any square matrix.
- ( ) It is only applicable to matrices with real entries.

**Nice work**

Correct! Eigendecomposition expresses a matrix in terms of its eigenvalues and eigenvectors.

**18.** What property do eigenvectors of a symmetric matrix have? 1 / 1 point

- ( ) They are always equal to the eigenvalues.
- ( ) They have the same magnitude.
- (•) They are orthogonal to each other.
- ( ) They are always parallel to each other.

**Nice work**

Correct! Eigenvectors of a symmetric matrix are orthogonal to each other.

**19.** What is the purpose of eigendecomposition? 1 / 1 point

- (•) To express a matrix in terms of its eigenvalues and eigenvectors.
- ( ) To find the inverse of a matrix.
- ( ) To determine the rank of a matrix.
- ( ) To calculate the trace of a matrix.

**Nice work**

Correct! Eigendecomposition expresses a matrix in terms of its eigenvalues and eigenvectors, simplifying many operations.

**20.** What is the identity matrix used for in the context of eigenvalues? 1 / 1 point

- ( ) To calculate the inverse of a matrix.
- (•) To form the equation $\det(A - \lambda I) = 0$ for finding eigenvalues.
- ( ) To determine if a matrix is symmetric.
- ( ) To scale eigenvectors.

**Nice work**

Correct! The identity matrix is used in the characteristic equation to find eigenvalues.

**21.** What is the primary purpose of using eigendecomposition in data analysis? 1 / 1 point

- ( ) To increase the number of dimensions in a dataset.
- ( ) To ensure all vectors are parallel.
- ( ) To convert all data into Fahrenheit.
- (•) To identify orthogonal and compact representations of vector space.

**Nice work**

Correct! Eigendecomposition helps in identifying orthogonal and compact representations of vector space, which is crucial for accurate data analysis.

**22.** What does PCA primarily aim to achieve in data analysis? 1 / 1 point

- ( ) To ensure all data points are equidistant.
- ( ) To increase the correlation between data dimensions.
- (•) To find the eigenvalues and eigenvectors of the covariance matrix.
- ( ) To convert data into a four-dimensional vector space.

**Nice work**

Correct! PCA aims to find the eigenvalues and eigenvectors of the covariance matrix to identify principal components.

**23.** What is the role of eigenvalues in PCA? 1 / 1 point

- ( ) They determine the redundancy of dimensions.
- (•) They indicate the variance captured by each principal component.
- ( ) They ensure all dimensions are orthogonal.
- ( ) They are used to convert data into Fahrenheit.

**Nice work**

Correct! Eigenvalues indicate the amount of variance captured by each principal component.

**24.** Which of the following is a feature extraction technique? 1 / 1 point

- (•) Principal Component Analysis (PCA)
- ( ) Model-based selection
- ( ) Euclidean distance
- ( ) Statistical tests

**Nice work**

Correct! PCA is a feature extraction technique that creates new variables by combining existing ones.

**25.** What is the key difference between PCA and SVD? 1 / 1 point

- ( ) PCA operates on the original data matrix, while SVD operates on the covariance matrix.
- ( ) PCA is used for structured data, while SVD is used for unstructured data.
- ( ) PCA is more numerically stable than SVD.
- (•) SVD can be used for non-square matrices, while PCA requires square matrices.

**Nice work**

Correct! SVD can handle non-square matrices, while PCA requires a square covariance matrix.